

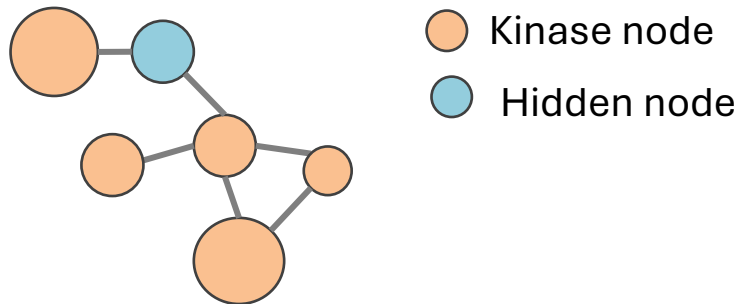
Kinase networks and pathways

Interpretation guide



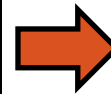
How are networks generated?

Networks are generated from high scoring* kinases. Networks connect key kinases through high-confidence links and may include connector (hidden) nodes which are not in the data.

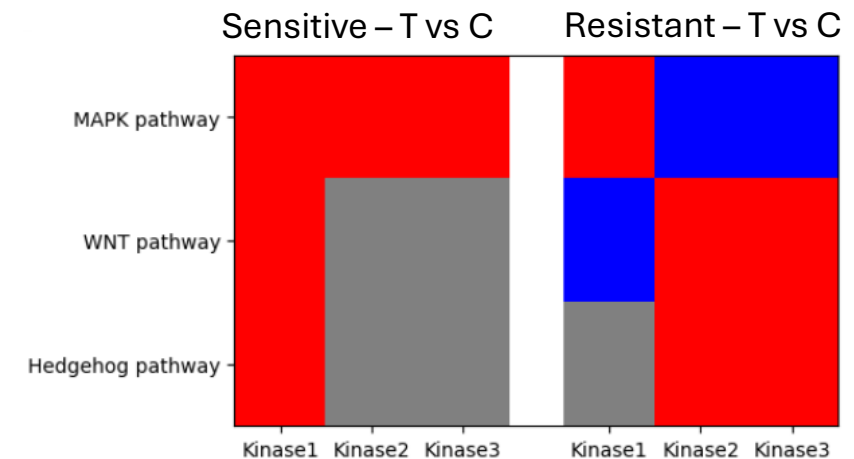


Files:

Network.html: an interactive network figure which shows the LogFC (Kinase Statistic) and the pathways. Nodes and pathways are searchable. Networks help to understand kinases relationships.



Network nodes are annotated with pathways by EnrichR, using the [Reactome](#) db.



Files:

Kinase_pathway_heatmaps folder: start exploring pathways with these figures to assess the change in the pathways across comparisons.

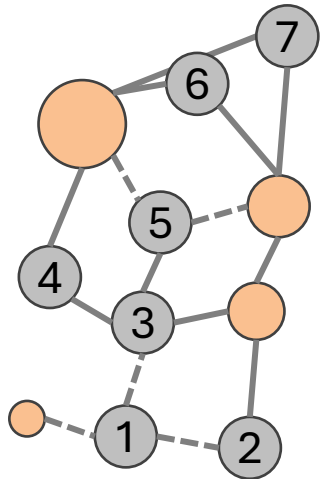
***Params.csv:** shows how the data was pre-filtered. spec_cutoff is the threshold for specificity score. perc_cutoffs are the threshold(s) for percentile, which is usually 0.01 (shown at the end of the filename as _p01) so everything is used above the specificity cutoff.

Network generation enhances UKA results



INPUTS:

- STRING as reference network;
- selected nodes from UKA



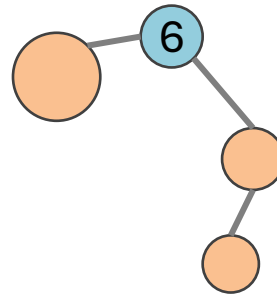
- Nodes
- Input nodes (size ~ LFC)

- High evidence edge
- - - Low evidence edge



RESULT:

“OPTIMAL” SUBNETWORK



- **Key kinases & high-confidence links**
 - Less FP (false positive – weakly connected)
 - Less FN (false negative, hidden nodes)
- Hidden nodes



Generating networks enhances robustness of UKA results:

- pruning weakly connected or low-fold-change nodes (probably FP)
- and recovering missing connections through hidden nodes (probably FN).

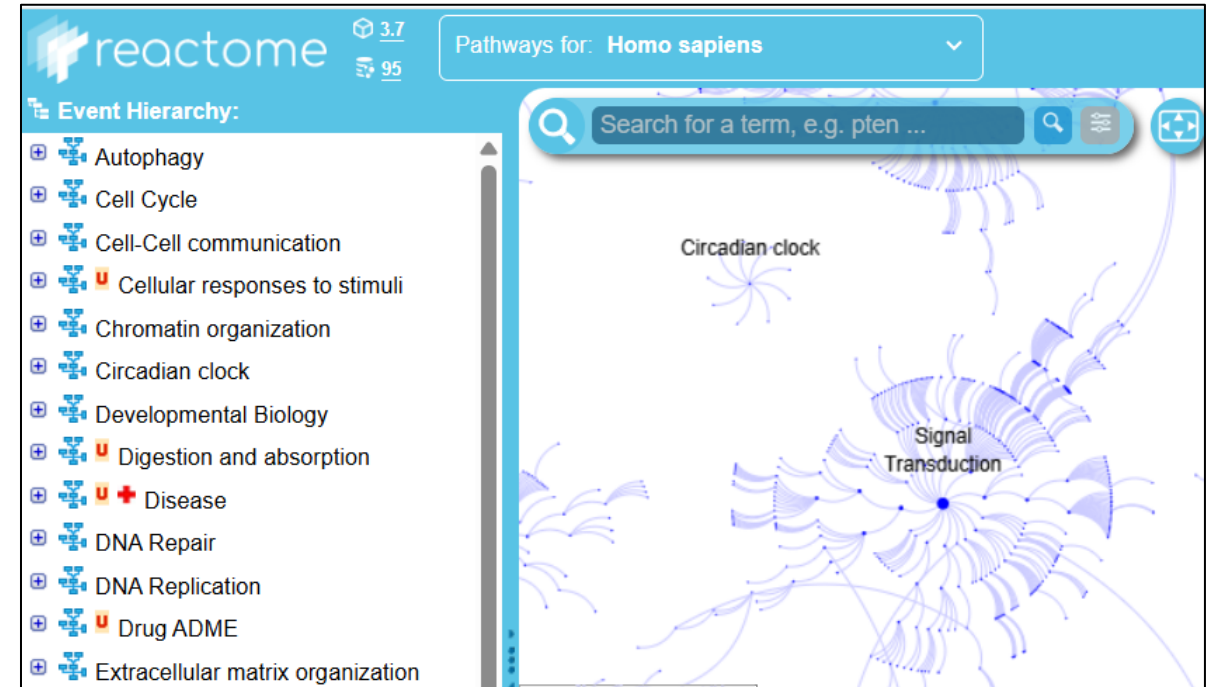
The Prize-Collecting Steiner Forest (PCSF) algorithm, given the reference network and input nodes, constructs a subnetwork through optimizing to:

- include more input nodes with high prize (prize = absolute LFC)
- Decrease edge costs (low cost ~ high evidence).

How to select most relevant pathways?

Use the Pathways_<T vs C>.csv:

- Every pathway belongs to a Reactome pathway group (e.g. AKT signaling is under Signal Transduction).
- These columns can help to identify relevant pathways:
 - *N_hits*, *pathway_size*, *overlap columns* show how many pathway members are in the data. (e.g. *n_hits* = 3, *pathway_size* = 10 means 3 kinases from the results are found in a pathway of 10 members).
 - *Hierarchy_n column*: shows the hierarchy level (e.g. Signal Transduction is level 1). Top levels are generic, bottom levels are specific. Middle levels might be the most interesting.



The **Pathways_<T vs C>.csv** file lists the selected pathways for each gene cluster (column Genes). However, additional pathways may also be annotated to the same gene cluster. In this file, the pathway chosen is the highest in the hierarchy (i.e., the most general). If a pathway is irrelevant, use **Pathways_<T vs C>.all.csv** to check whether there is another, more relevant pathway for the same gene cluster.

How to select most relevant pathways?



Which are the most relevant pathways?

There is no simple metric – p values alone are not enough to choose the most relevant ones. The reasons:

- Pathways usually share members, and a pathway that is relevant in one model may be irrelevant in another.
- statistical tests do not consider how specific members are to a pathway. As a result, a pathway may receive a strong p-value even if only non-specific members are detected.

→ Therefore, we recommend assessing pathway relevance using biological domain knowledge rather than relying on p-values or combined scores.

- **Nodes.csv:** list of all kinase and hidden nodes in the network
- **Nodes_clusters:** Network clusters.
- **Missing nodes.csv:** kinases that passed the thresholds but are not in the network because they are weakly connected / low fold change. Might be false positives.
- **Nodes with pathways.csv:** nodes to pathways mapping
- Folders with the files of the network html (needed for the html to work. Do not delete it.)

The PCSF ([Prize-Collecting Steiner Forest algorithm](#)) solves a network optimization problem. Given a reference network, where the nodes have prizes and the edges have costs, it aims to identify a subnetwork with a given node input. Subnetwork construction is an optimization to minimize the total edge costs and the total node prizes left out:

$$\text{Minimize} \left(\sum \text{edge cost} + b \sum \text{prize of nodes left out} \right)$$

Where b is used to tune the prizes of nodes relative to edge costs. Higher b leads to more input nodes incorporated in the result subnetwork.

We use the PCSF R package with the STRING network (v12.0) containing all edges as the reference interactome. Node prizes are assigned based on the UKA and Target Fold Changes.

Evidence channels of STRING



The STRING protein-protein interaction database contains interactions from the following sources:

Evidence Channel	Type	Reliability	Key Strength
Experimental	Direct	 Highest	Proof of direct physical or genetic interaction.
Curated Databases	Direct	 Very High	Expert-validated functional association in pathways/complexes.
Gene Fusion	Genomic	 High	Strong evolutionary evidence for functional interaction.
Gene Neighborhood	Genomic	 High (Prok) / Low (Euk)	Evidence for operons and co-regulation (mainly in bacteria).
Gene Co-occurrence	Genomic	 Medium-High	Evidence for participation in the same pathway/system.
Co-expression	Functional	 Medium	Strong evidence for co-regulation and functional association.
Text mining	Literature	 Lower	High-level associative